

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
Faculty Name	BALAJIAH GUDIPUDI
Student Name	PALAVALASA RAHUL
Roll Number	2520090016
Branch	CSIT
Course Outcome	1

Case Study Topic: Discord Server Roster – Fast "Find the k-th Most Senior Member"
Query using AVL Tree with Rank Augmentation

Word Done Summary:

In this case study, an AVL Tree with Rank Augmentation was implemented to efficiently answer the query: "Who is the k-th most senior member in a Discord server?" The Discord server contains approximately 200,000 members, where each member is identified by a unique join timestamp. The oldest member is considered the most senior. A normal AVL Tree provides efficient insertion and deletion operations in $O(\log n)$ time. However, finding the k-th senior member requires traversal of multiple nodes, leading to $O(n)$ complexity in the worst case. To improve query performance, the AVL Tree was augmented by storing the subtree size at every node. This additional information allows rank-based queries to be answered in $O(\log n)$ time.

Implementation Details:

1. AVL Tree node creation.
2. Height maintenance.
3. Size augmentation at each node.
4. AVL balancing through rotations (LL, RR, LR, RL).
5. Insertion operation.
6. k-th smallest (most senior) member retrieval.
7. In-order traversal for verification.
8. Complexity analysis and performance evaluation.

Result:

All timestamps were inserted into the AVL Tree AVL balance factors were maintained. Rotations were performed whenever required. Sub tree sizes were updated.

Output:

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Inorder Traversal:
1001 1002 1003 1004 1005 1006 1007 1008 1010 1012

3-th most senior member is: 1003
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GitHub Link:

<https://github.com/palavalasarahul-blip/DSA-CASE-STUDIES>

Student Signature	Faculty Signature
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